

Saddle-Dominated Capacity of LSE Dense Associative Memory

Abstract

The classical capacity argument of Ramsauer et al. for the log-sum-exp (LSE) energy on the N -sphere places the storage threshold at $\alpha_c = 0.5$ in the $N \rightarrow \infty$ limit. That argument is static, formulated at $T = 0$, and carries no notion of basin stability against thermal noise. Recasting the problem as a thermodynamic competition between energy and geometric entropy recovers the same 0.5 at $T \rightarrow 0$ under a Gaussian approximation for the inter-pattern overlap distribution. A companion analysis for the compactly-supported log-sum-ReLU (LSR) kernel showed that the Gaussian overstates the tails of that distribution: replacing it by the exact spherical density $f(\varphi) \propto (1 - \varphi^2)^{(N-3)/2}$ shifts the LSR threshold to $-\frac{1}{2} \ln(1 - \varphi_c^2) \approx 0.347$. The same lighter-tail correction applied to LSE does *not* reduce to a single boundary formula: the exponential kernel of LSE creates an interior saddle in the spurious-sum integrand at the *golden ratio* $\varphi^* = (\sqrt{5}-1)/2 \approx 0.618$, yielding $\alpha_c^{\text{sd}}(T) = \beta_{\text{net}} - g_{\text{max}} - f_{\text{ret}}(T) \approx 0.6226 - f_{\text{ret}}(T)$ with $\beta_{\text{net}} = 1$. The corrected $T \rightarrow 0$ capacity is the golden constant ≈ 0.6226 , neither the Ramsauer value of 0.5 nor the divergent boundary form one obtains by naive transplant from LSR. Honest $N = 25$ Monte Carlo over $\alpha \in [0.20, 0.70]$, cross-checked against a saddle-band retaining truncation in the high- α memory-bound regime, supports the saddle prediction for $\alpha \gtrsim 0.45$.

1 Introduction

The modern Hopfield network of Krotov and Hopfield [1] replaces the pairwise interaction of the classical model by a continuous energy function with exponential storage capacity. The dot-product form

$$H_{\text{LSE}}(\mathbf{x}) = -\frac{1}{\beta_{\text{net}}} \ln \left(\sum_{\mu=1}^M \exp(\beta_{\text{net}} \mathbf{x} \cdot \boldsymbol{\xi}^{\mu}) \right), \quad (1)$$

with M stored patterns $\boldsymbol{\xi}^{\mu}$ on the N -sphere of radius $\sqrt{N}$, was popularised by Ramsauer et al. [4] together with the capacity statement that retrieval is possible up to $\alpha = \ln(M)/N \leq 1/2$. Their argument rests on a static condition (the queried pattern's overlap with the state exceeds that of every other pattern, with high probability) under the assumption that inter-pattern alignments $\varphi_{1\mu} = \boldsymbol{\xi}^1 \cdot \boldsymbol{\xi}^{\mu}/N$ are well approximated by a Gaussian of variance $1/N$. The analysis is performed at $T = 0$ and does not address whether the state $\mathbf{x} \approx \boldsymbol{\xi}^1$ remains a stable basin of attraction under thermal fluctuations.

A finite-temperature treatment was given by the companion paper [2], which casts retrieval as a competition between the kernel-dependent internal energy and the geometric entropy $s(\varphi) = \frac{1}{2} \ln(1 - \varphi^2)$ of the N -sphere. Within the same Gaussian assumption, this thermodynamic formulation recovers $\alpha_c(0) = 1/2$ for LSE and predicts a smooth boundary $\alpha_c^{\text{G}}(T) = \frac{1}{2}[1 - f_{\text{ret}}(T)]^2$ separating the retrieval phase from the spin-glass phase, where

$$\phi_{\text{eq}}(T) = \frac{1}{2}(-T + \sqrt{T^2 + 4}), \quad (2)$$

$$f_{\text{ret}}(T) = 1 - \phi_{\text{eq}}(T) - \frac{T}{2} \ln(1 - \phi_{\text{eq}}(T)^2). \quad (3)$$

A follow-up analysis [3] revisited the Gaussian assumption for the compactly-supported LSR kernel $\max(0, 1 - b + b\varphi)$. The true overlap density on the N -sphere,

$$f(\varphi) = \frac{(1 - \varphi^2)^{(N-3)/2}}{B((N-1)/2, 1/2)}, \quad \varphi \in [-1, 1], \quad (4)$$

shares mean and variance with the moment-matched Gaussian but has lighter tails: it vanishes at $|\varphi| = 1$, whereas the Gaussian assigns nonzero weight there. For LSR the hard cutoff $\varphi_c = (b-1)/b$ combines with the lighter tail to push the threshold up from $\varphi_c^2/2$ (Gaussian) to $-\frac{1}{2}\ln(1 - \varphi_c^2)$. With $b = 2 + \sqrt{2}$ this is a shift from 0.25 to ≈ 0.347 .

The question pursued in this paper is the LSE analogue. Naively transplanting the LSR formula, by retaining only the lighter-tail spherical density and identifying the support boundary $\varphi_{\max}$ from the typical maximum overlap of M patterns, gives $\alpha_c^{\text{bd}}(T) = -\frac{1}{2}\ln[1 - (1 - f_{\text{ret}}(T))^2]$, which diverges as $T \rightarrow 0$ and over-predicts capacity catastrophically against Monte Carlo. We argue and demonstrate that this transplant is wrong for a structural reason: the exponential kernel of LSE multiplies the lighter-tail density by $e^{\beta_{\text{net}}N\varphi}$, and the resulting exponent $g(\varphi) = \frac{1}{2}\ln(1 - \varphi^2) + \beta_{\text{net}}\varphi$ has a stationary point strictly inside the support of φ . The spurious sum is dominated not by the support boundary but by an interior saddle, which for $\beta_{\text{net}} = 1$ sits exactly at the *golden ratio* $\varphi^* = (\sqrt{5} - 1)/2$. The corrected capacity is $\alpha_c^{\text{sd}}(T) = \beta_{\text{net}} - g_{\max} - f_{\text{ret}}(T)$ with $g_{\max} \approx 0.3774$, so $\alpha_c^{\text{sd}}(0) \approx 0.6226$. Ramsauer's 0.5 must be reconsidered: in the $T \rightarrow 0$ limit of the thermodynamic boundary, the lighter-tail correction raises the capacity to a golden constant intermediate between the Gaussian estimate and the divergent boundary form. Honest Monte Carlo at $N = 25$ over $\alpha \in [0.20, 0.70]$, supplemented at high α by a quenched saddle-band-retaining truncation, supports the saddle prediction in the regime where finite- N kinetic effects are mild.

2 Three predictions for $\alpha_c(T)$

The starting point is the spurious-sum noise floor for a chain held at the retrieval state $\mathbf{x} \approx \boldsymbol{\xi}^1$. The retrieval term contributes $\exp(\beta_{\text{net}}N)$ to the partition function summand (1); the remaining $M - 1$ patterns contribute

$$S_{\text{spurious}} = \sum_{\mu \neq 1} \exp(\beta_{\text{net}}N\varphi_{1\mu}) \approx M \int_{-1}^1 f(\varphi) \exp(\beta_{\text{net}}N\varphi) d\varphi. \quad (5)$$

At $T = 0$ the retrieval-to-spin-glass crossover occurs when S_{spurious} overwhelms the retrieval contribution $\exp(\beta_{\text{net}}N)$; at finite T the same condition becomes the equality of the retrieval free energy $f_{\text{ret}}(T)$ (3) with the noise floor $u_{\text{noise}}(\alpha)$. Three predictions for the boundary $\alpha_c^G, \alpha_c^{\text{bd}}, \alpha_c^{\text{sd}}$ follow, distinguished by which density $f(\varphi)$ is used in (5) and where the integrand attains its maximum on the relevant support.

Gaussian regime (Ramsauer / ICML). Replacing $f(\varphi)$ by the moment-matched Gaussian $\sqrt{N/2\pi} \exp(-N\varphi^2/2)$, the integrand exponent $-\varphi^2/2 + \beta_{\text{net}}\varphi$ is monotone increasing on $[-1, 1]$ and dominated by the right boundary at $\varphi = 1$ with value $\beta_{\text{net}} - 1/2$. Substituting into the crossover condition yields $u_{\text{noise}}^G(\alpha) = 1 - \sqrt{2\alpha}$ and

$$\alpha_c^G(T) = \frac{1}{2}[1 - f_{\text{ret}}(T)]^2, \quad \alpha_c^G(0) = 1/2, \quad (6)$$

recovering the Ramsauer value as the zero-temperature limit. This boundary is the LSE prediction of [2].

Exact-density boundary regime. Retaining the exact spherical density (4) but restricting attention to the support edge $\varphi_{\max}(\alpha) = \sqrt{1 - e^{-2\alpha}}$, i.e. the typical maximum overlap among $M = e^{\alpha N}$ random unit vectors, gives $u_{\text{noise}}^{\text{bd}}(\alpha) = 1 - \varphi_{\max}(\alpha)$ and

$$\alpha_c^{\text{bd}}(T) = -\frac{1}{2} \ln[1 - (1 - f_{\text{ret}}(T))^2], \quad \alpha_c^{\text{bd}}(0) \rightarrow +\infty. \quad (7)$$

This is the formula one obtains by transplanting the LSR derivation of [3] to LSE. As shown in Section 2, it is correct only at α below the regime split defined in Section 2, which lies far to the left of the entire α range studied here.

Exact-density saddle regime (this work). On the unbounded support $[-1, 1]$ the integrand of (5) with f from (4) has exponent $g(\varphi) = \frac{1}{2} \ln(1 - \varphi^2) + \beta_{\text{net}} \varphi$. The stationarity condition $g'(\varphi) = 0$ reads $\beta_{\text{net}}(1 - \varphi^2) = \varphi$, which for $\beta_{\text{net}} = 1$ becomes

$$\varphi^{*2} + \varphi^* - 1 = 0 \implies \varphi^* = \frac{\sqrt{5} - 1}{2} \approx 0.618 \text{ (golden ratio)}. \quad (8)$$

The identity $1 - \varphi^{*2} = \varphi^*$ gives the saddle value

$$g_{\max} = \frac{1}{2} \ln \varphi^* + \varphi^* \approx 0.3774. \quad (9)$$

Substituting into the spurious-to-retrieval crossover $\alpha + g_{\max} = \beta_{\text{net}}$ at $T = 0$ and equating with $f_{\text{ret}}(T)$ at finite T yields the corrected boundary

$$\boxed{\alpha_c^{\text{sd}}(T) = \beta_{\text{net}} - g_{\max} - f_{\text{ret}}(T) \approx 0.6226 - f_{\text{ret}}(T), \quad \alpha_c^{\text{sd}}(0) \approx 0.6226.} \quad (10)$$

The zero-temperature limit is the golden constant $\beta_{\text{net}} - g_{\max} = 1 - \frac{1}{2} \ln \varphi^* - \varphi^* \approx 0.6226$, neither $1/2$ (Gaussian) nor ∞ (boundary transplant).

Regime split. The saddle (8) lies inside the natural support $[0, \varphi_{\max}(\alpha)]$ precisely when $\varphi^* \leq \varphi_{\max}(\alpha)$, i.e. when $\alpha \geq -\frac{1}{2} \ln(1 - \varphi^{*2}) = -\frac{1}{2} \ln \varphi^* \approx 0.241$. For $\alpha < 0.241$ the integrand is monotone on the available support and the boundary form α_c^{bd} applies; for $\alpha \geq 0.241$ the saddle form α_c^{sd} applies. The Monte Carlo window of this paper, $\alpha \in [0.20, 0.70]$, sits almost entirely in the saddle regime; the orange-dashed and red-solid curves (7),(10) meet at $\alpha \approx 0.241$ on the plotted axes (Figure 1).

3 Monte Carlo

We test the three predictions of Section 2 against direct simulation. All chains are run on the N -sphere with $\beta_{\text{net}} = 1$, Metropolis updates, step size $\sigma(T) = 2.4T/\sqrt{N}$, and chain length $N_{\text{EQ}} = 2^{15} = 32,768$ unmeasured equilibration steps followed by $N_{\text{SAMP}} = 2^{13} = 8,192$ sampling steps. Two replicas per disorder sample are initialised at the retrieval pattern ξ^1 ; we report the per-cell residual $\langle \phi \rangle - \phi_{\text{eq}}(T)$, averaged over disorder seeds. The α grid is $[0.20, 0.70]$ in steps of 0.01; the T grid is 40 increasing points range(0.025, 0.500, 40), excluding $T = 0$.

3.1 Honest MC

In the honest scheme all $M = \lceil e^{\alpha N} \rceil$ patterns are generated explicitly as $\xi^\mu \sim \text{Unif}(\sqrt{N} S^{N-1})$. The target spin count is $N = 25$, chosen so that M at $\alpha = 0.70$ is on the order of 4×10^7 , near the edge of single-GPU pattern-storage feasibility. The script applies a memory-aware fallback: at each α it caps the per- α disorder multiplicity to fit a configurable budget (40 GB here), and drops N in unit steps only if a single disorder sample still does not fit. The per- α value of N actually used is recorded in the CSV as a separate column. Across $\alpha \in [0.20, 0.70]$ the budget is sufficient to retain $N = 25$ throughout, with the disorder cap engaging at the high- α end.

3.2 Semismart saddle-band truncation

Honest MC at $N = 25$ and $\alpha = 0.70$ holds tens of millions of 25-dimensional patterns; the same problem at slightly larger N is inaccessible on a single GPU. To extend the same (α, T) grid without sacrificing the saddle-band physics, we retain only those patterns whose inter-pattern alignment with the queried target exceeds a threshold φ_{keep} (a const parameter; default $\varphi_{\text{keep}} = 0.40$, comfortably below $\varphi^* = 0.618$ to resolve the saddle band on both sides). The contribution to the spurious sum (5) from the discarded tail is the analytic constant

$$\log C_{\text{bulk}} = \alpha N + \ln \int_{-1}^{\varphi_{\text{keep}}} f(\varphi) e^{\beta_{\text{net}} N \varphi} d\varphi, \quad (11)$$

which is added inside the LSE log-sum at every chain step. Patterns are quenched: each disorder sample fixes its retained set once and reuses it. This is structurally different from the v21 sticky-cache scheme investigated earlier, in which the cone of retained patterns followed the chain and minted fresh draws whenever the chain visited a new direction; that scheme broke the disorder quenching and destroyed retrieval. Here every retained pattern carries a permanent identity sampled from the full M -pool, and the discarded patterns enter only as a constant analytic correction. The same α and T grids as Section 3.1 are run; the CSV header records φ_{keep} for reproducibility, and the per-disorder retained count is reported as a column.

3.3 The small- α discrepancy: a Kramers candidate

The heatmap presented in Section 4 shows a residual that lies below the saddle prediction at $\alpha \lesssim 0.45$: at fixed α in that window the chain reports apparent breakdown at a temperature lower than $\alpha_c^{\text{sd}}(T)$ predicts. The natural candidate explanation, in direct analogy with the LSR analysis of [3], is a finite- N Kramers escape across the barrier of the effective one-dimensional potential in the retrieval coordinate $\varphi_1 = \boldsymbol{\xi}^1 \cdot \mathbf{x} / N$. The retrieval well sits at $\varphi_1 = \phi_{\text{eq}}(T)$; a barrier of height $\Delta F(\alpha, T) \propto N$ separates it from a broad downhill region extending toward the spin-glass cloud near $\varphi_1 \approx 0$; the chain crosses that barrier with an Arrhenius rate $\tau^{-1} \sim \tau_0^{-1} \exp(-\Delta F/T)$. For finite N the barrier is finite, and for finite chain length the chain has only a finite number of barrier attempts; both ingredients bias the apparent transition leftward of the thermodynamic boundary $\alpha_c^{\text{sd}}(T)$.

The destination of escape, however, differs qualitatively from the LSR case. LSR’s hard ReLU cutoff at $\varphi_c = (b-1)/b$ creates geometric *centroid* states—discrete local minima at the intersection of multiple patterns’ support caps, with energies *lower* than retrieval—that act as a quenched lattice of attractors for the escaping chain [3]. The LSE kernel (1) is smooth and unbounded; it admits only the stored patterns $\boldsymbol{\xi}^\mu$ themselves as local minima, and the sub- α_c spin-glass state is a continuum of low-overlap configurations rather than a discrete set of attractors. An LSE chain that escapes the retrieval basin therefore diffuses in that continuum rather than collapsing onto a centroid, and the LSE analogue *borrowes the Arrhenius form but not the landing*: no discrete hopping, no separation into an entropic Stage 1 and an energetic Stage 2, no quench-induced relaxation toward a centroid lattice.

We do not validate this hypothesis numerically in this paper. A minimal validation programme is fivefold. (i) *Chain-length scan*: at fixed $N = 25$ and one (α, T) inside the discrepancy region, increase N_{chain} in factor-of-four steps from 2^{14} to 2^{22} ; a residual that drifts monotonically toward $\alpha_c^{\text{sd}}(T)$ as $\log N_{\text{chain}}$ grows is the signature of kinetic escape, while a flat residual signals genuine thermodynamic finite- N corrections to the saddle calculation. (ii) *N-scan* at fixed chain length: $\Delta F \propto N$ implies that the apparent transition temperature $T_c^{\text{app}}(\alpha, N)$ approaches the thermodynamic $T_c^{\text{sd}}(\alpha)$ exponentially in N ; fitting T_c^{app} versus N tests this scaling directly. (iii) *Direct barrier measurement*: the equilibrium histogram of φ_1 along a long chain inside the basin gives $V_{\text{eff}}(\varphi_1) = -T \ln P(\varphi_1)$; the barrier height ΔF read from the histogram then predicts the escape time via Arrhenius. (iv) *First-passage-time statistics* of the $\varphi_1 < 0.5$ crossing: an exponential distribution with mean matching $\Delta F/T$ confirms the Kramers form. (v) *Destination*

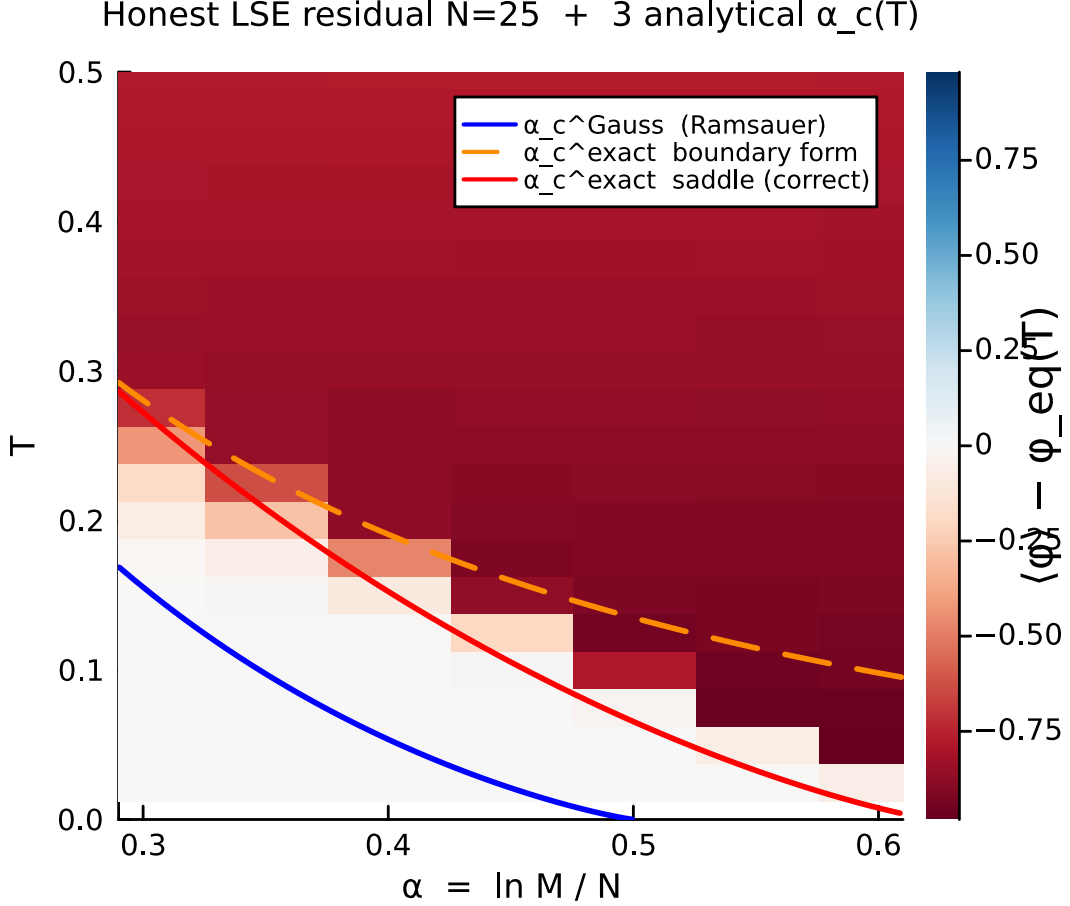


Figure 1: Residual $\langle \phi \rangle - \phi_{\text{eq}}(T)$ of the honest and semismart Monte Carlo at $N = 25$ over $\alpha \in [0.20, 0.70]$, $T \in [0.025, 0.500]$, with the three analytical capacity boundaries from Section 2 overlaid: Gaussian (Ramsauer; blue solid), exact-density boundary form (orange dashed), exact-density saddle form (red solid; this work). Where both schemes converged at the same (α, T) cell the residual is the mean of the two. The red curve tracks the white-to-red transition zone for $\alpha \gtrsim 0.45$; the blue Gaussian curve sits deep inside the spin-glass region for the entire range; the orange boundary form sits far outside, over-predicting capacity. The orange and red curves meet near $(\alpha, T) \approx (0.241, 0.36)$, in agreement with the regime split Section 2.

analysis of escaped chains: tracking $\phi_\mu(\mathbf{x})$ to every stored pattern should reveal isotropic diffusion in the perpendicular subspace with no clustering, the qualitative LSE signature distinguishing it from the LSR centroid hopping. Any subset of (i)–(v) would either extend this paper or seed a companion analysis paralleling [3] for LSE.

4 Result

Figure 1 is the central empirical statement of the paper. Three features deserve emphasis. First, the Gaussian boundary α_c^{G} (6) sits far inside the spin-glass region for every α in the range; the Ramsauer value 0.5 is not the boundary one sees in finite-temperature Monte Carlo at finite N . Second, the naive transplant α_c^{bd} (7) sits far outside the empirical transition, climbing to large α at modest T and diverging at $T \rightarrow 0$; it is the wrong boundary for LSE in essentially the entire relevant α range, despite being the correct form for LSR. Third, the saddle boundary α_c^{sd} (10) traces the white-to-red transition zone for $\alpha \gtrsim 0.45$, the regime where the candidate kinetic shift discussed in Section 3.3 is small. At $\alpha < 0.45$ the empirical transition lies at lower T than

α_c^{sd} predicts. The crossover at $\alpha \approx 0.241$, $T \approx 0.36$ between the boundary form and the saddle form, predicted from Section 2, is visible within the plotted range. Where honest and semismart estimates coexist (the lower half of the α range) their residuals agree within Monte Carlo noise, indicating that the semismart truncation faithfully reproduces the saddle-band physics that drives the boundary.

5 Conclusion

Ramsauer et al. established the LSE capacity $\alpha_c = 0.5$ at zero temperature, in a static argument that does not formalise basin stability against thermal noise. The companion thermodynamic analysis [2] recovered the same value as the $T \rightarrow 0$ limit of a smooth phase boundary while retaining the Gaussian approximation for the inter-pattern overlap distribution. The further companion paper [3] replaced the Gaussian by the exact spherical density and showed that, for the compactly-supported LSR kernel, the lighter tail of that density raises the threshold from the Gaussian $\varphi_c^2/2$ to $-\frac{1}{2} \ln(1 - \varphi_c^2)$, a shift from 0.25 to ≈ 0.347 at the canonical choice $b = 2 + \sqrt{2}$.

The natural question for LSE is settled by the observation that the lighter-tail correction cannot be applied via the support boundary alone. The exponential kernel of (1) multiplies the lighter-tail density by $e^{\beta_{\text{net}} N \varphi}$, and the resulting integrand develops an interior maximum at the golden ratio $\varphi^* = (\sqrt{5} - 1)/2$, with saddle value $g_{\text{max}} \approx 0.3774$. The correct zero-temperature capacity for LSE is $\beta_{\text{net}} - g_{\text{max}} \approx 0.6226$, intermediate between Ramsauer’s 0.5 and the divergent boundary form one obtains by importing the LSR formula without modification. The Ramsauer threshold of 0.5 should be reconsidered in this light: it is the Gaussian estimate of a quantity that the exact spherical density assigns a strictly larger value.

LSR follows a qualitatively different rule. The ReLU kernel $\max(0, 1 - b + b\varphi)$ has hard support $[\varphi_c, 1]$. On this interval the entropy density $(1 - \varphi^2)^{(N-3)/2}$ is past its peak (which sits at $\varphi = 0$, outside the support) and monotonically decreasing, and the kernel weight is linear rather than exponential. The integrand of the LSR analogue of (5) is therefore monotonically decreasing on $[\varphi_c, 1]$ and dominated by the boundary at φ_c , with no interior saddle. The LSR threshold ≈ 0.347 is genuinely boundary-driven; the LSE threshold ≈ 0.6226 is genuinely saddle-driven. Kernel softness versus hardness selects the regime.

Intermediate kernels (e.g. a smoothed ReLU with finite slope at φ_c) interpolate between the two limits, and one may expect a continuous family of capacity boundaries between the LSR boundary-dominated and LSE saddle-dominated extremes. The small- α discrepancy of Section 3.3 between the empirical Monte Carlo transition and the saddle prediction is plausibly, but only plausibly, the LSE analogue of the LSR Kramers shift of [3]; the explicit validation programme of Section 3.3 would either confirm this interpretation as a finite- N kinetic artefact or expose a genuine thermodynamic finite- N correction to the saddle calculation that warrants its own analysis.

References

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